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Title: Stock Market Direction Prediction Using Machine Learning Classification: A Comparative Analysis of Supervised Classifiers on SPY ETF Data

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Abstract

Stock market direction prediction remains a longstanding challenge in financial research, particularly considering the Efficient Market Hypothesis, which suggests that historical market data should carry no systematic predictive signal. While prior studies have tested machine learning models on individual stocks and market indices, limited evidence exists on broad market ETFs evaluated across multiple distinct crisis regimes. This study asks: How effectively can supervised machine learning classification algorithms predict next-day price direction of the SPY ETF using historical price-based technical indicators? Using daily SPY OHLCV data from 2005 to 2024, four supervised learning classifiers were evaluated, namely Logistic Regression, Random Forest, Gradient Boosting, and XGBoost, across three iterative feature configurations using a chronological 70/30 train/test split. Across all iterations, every model performed below the naive baseline of 54.4%, with AUC-ROC values consistently ranging from 0.48 to 0.51. A secondary finding was that feature quality mattered more than feature quantity, as expanding the feature set did not improve performance, while a refined momentum-focused configuration produced the strongest results. Overall, the findings support weak-form EMH and suggest that next-day direction prediction of a broad market ETF lies beyond the reach of historical price data alone.

1. Introduction

Stock market prediction remains one of the most difficult and consequential problems in financial research because even marginal improvements in forecasting accuracy can carry substantial economic value. This challenge is especially relevant for SPY, the exchange-traded fund that tracks the S&P 500, as it provides a highly liquid and commonly used benchmark for the broader US market. In such an environment, traditional rule-based analysis and human judgement face clear limitations. Therefore, this study applies machine learning classification to evaluate whether directional patterns in historical SPY data carry a predictive signal, or whether the market's efficiency has already absorbed them.

However, any attempt to predict price direction from historical data immediately encounters a fundamental theoretical challenge due to the Efficient Market Hypothesis (EMH). Fama (1970) argues that in a weak-form efficient market, prices fully reflect all historical price and volume information, implying that excess returns cannot be consistently generated through technical analysis alone. In this context, low predictive accuracy should not be automatically interpreted as a methodological failure. Rather, it becomes an empirical test of whether weak-form efficiency holds for SPY: near-random model performance would support EMH, whereas consistent outperformance would challenge it.

This interpretation is challenged by the Adaptive Market Hypothesis (AMH), which suggests that market efficiency is not fixed but varies across time and conditions (Butler and Kazakov, 2012). This motivates the present study's chronological design, which trains models on data encompassing the 2008 financial crisis and tests them on the COVID-19 shock and post-pandemic inflationary period. This allows the study to assess not only overall predictive accuracy, but whether models trained on one crisis regime can generalise to a structurally different one.

Existing studies have largely focused on individual stocks or non-ETF indices, such as Kumar et al. (2018), Jiao and Jakubowicz (2017), and Zouaghia et al. (2023). This leaves limited evidence on broad market ETFs, despite Batool and Fatima's (2025) suggestion that index-level markets are typically more efficient than individual stocks. Thus, this study asks: How effectively can supervised machine learning classification algorithms predict next-day price direction of the SPY ETF using historical price-based technical indicators?

2. Related Works

2.1 EMH, AMH, and the Theoretical Debate

The theoretical foundation for machine learning-based stock prediction lies in the tension between the Efficient Market Hypothesis (EMH) and later challenges to its fixed view of market behaviour. Fama (1970) argues that under weak-form efficiency, historical price and volume information is already reflected in current prices, implying that past market data should carry no systematic predictive signal. More recent literature has questioned whether this position fully holds in practice. Rouf et al. (2021) note that EMH historically dominated financial theory, but advances in machine learning and related predictive methods suggest that stock prices may be forecastable to a limited extent, indicating possible constraints on market efficiency.

This challenge is made more explicit by Butler and Kazakov (2012), who apply the Adaptive Market Hypothesis (AMH) to S&P 500 data from 2001 to 2010 and find that optimised technical indicators outperformed the market benchmark during specific sub-periods. This cyclical view of efficiency directly motivates the cross-regime design of the present study. Together, this debate motivates the empirical question of which algorithms and feature sets, if any, can capture residual predictive signal in historical market data in the context of a broad market ETF.

2.2 Feature Engineering and Technical Indicators

Prior literature consistently shows that feature selection plays a central role in stock market classification accuracy. Kumar et al. (2018) test 12 technical indicators, including moving averages, RSI, Williams %R, volatility, and CCI across Indian and US stocks, achieving accuracies of up to 72% and explicitly finding that reducing the number of indicators lowered performance across all models. Pardeshi and Kale (2021) similarly report that RSI, EMA, MACD, and price-volume analysis produced accuracies of roughly 75% on 5-day prediction windows for Indian stocks. Shashank et al. (2023) also find that six technical indicators, including RSI, Williams %R, and ATR, generated accuracies between 70% and 73%, while Sirohi et al. (2014) apply moving averages, the stochastic oscillator, and on-balance volume within a walk-forward framework to predict daily stock direction.

These studies directly informed the present paper's feature engineering strategy. The indicators appearing most frequently across prior work, particularly moving averages, RSI, MACD, stochastic measures, and volume-based signals, formed the basis of the initial and expanded feature sets tested. Nevertheless, Kumar et al.'s (2018) finding that additional indicators improve classification accuracy was not replicated. In this study, expanding the feature set in Iteration 2 led to a marginal decline in performance, suggesting that on a highly efficient instrument such as SPY, feature redundancy may matter more than feature quantity.

2.3 Algorithm Comparison and Benchmark Accuracy

Comparative studies generally find that ensemble methods outperform simpler classifiers in stock direction prediction tasks. Zouaghia et al. (2023) test five classifiers on NASDAQ data from 2018 to 2023, including periods covering the COVID-19 pandemic and the Russia-Ukraine war. They report that Random Forest achieved the highest accuracy at 61% using 20 features reduced through PCA. Akhilesh et al. (2024) similarly compare Support Vector Machine and Decision Tree models, finding that SVM reached 59.76% accuracy while Decision Tree achieved only 51.85%, a result close to random classification. Ashtagi et al. (2024) also find that Random Forest, with an accuracy of 65%, outperformed regression-based approaches when the task was framed as directional classification rather than price estimation. At the other extreme, Bareket and Pârv (2024) report accuracies of up to 97% when predicting medium-term 70-day index direction using constituent stock indicators across the Nasdaq 100, Dow Jones, and DAX.

Taken together, these studies suggest that both algorithm choice and prediction horizon materially shape reported performance. Bareket and Pârv's (2024) very high medium-term accuracy contrasts sharply with the near-random results often observed in next-day studies, implying that longer horizons allow patterns more time to emerge above short-term market noise. This makes next-day prediction the more demanding test of weak-form efficiency. For that reason, the most directly comparable evidence comes from Jiao and Jakubowicz's work on the S&P 500.

2.4 S&P 500 Evidence & Gap in Literature

Most directly, Jiao and Jakubowicz (2017) provide the closest comparison, applying four classifiers to 463 S&P 500 stocks from 2009 to 2017, using more than 200 technical indicators. Their results showed that self-prediction from historical price data alone produced test AUC values of approximately 0.5, indicating near-random classification performance. Predictive performance only improved meaningfully when external global market data was incorporated. This is reinforced by Batool and Fatima (2025), who argue that index-level markets such as the S&P 500 and NASDAQ are comparatively more efficient than individual stocks, implying that prediction at the ETF or index level is especially demanding.

Yet, existing studies have not specifically examined broad market ETFs across multiple distinct crisis regimes. No prior study has trained models on a period encompassing the 2008 financial crisis and evaluated them on the COVID-19 shock and post-pandemic inflationary environment, leaving the question of cross-regime generalisation unresolved. This study addresses that gap by applying machine learning classification to SPY, providing a cross-regime test of weak-form efficiency on a highly liquid broad market instrument.

3. Methodology

3.1 Dataset and Preprocessing

This study used daily SPY OHLCV data sourced from Kaggle and truncated to 2005-03-15 to 2024-04-15. The dataset contained 4,853 observations before cleaning and 4,804 after rows with missing values from rolling-window calculations were removed. SPY was selected over individual stocks to reduce company-specific noise and provide a cleaner test of broad market efficiency. Initial preprocessing, including date range selection and recalculation of baseline rolling variables, was completed before Python implementation to ensure consistency across the final dataset.

3.2 Target Variable

The prediction task was framed as binary classification, where Direction = 1 if Close(t+1) > Close(t), and 0 otherwise. The final class distribution was 2,611 up days (54.4%) and 2,193 down days (45.6%). This established a naive baseline of 54.4%, meaning any model below this threshold performs worse than always predicting an up day. Although the classes were moderately imbalanced, the distribution was not severe enough to require resampling techniques.

3.3 Feature Engineering

Feature engineering was conducted in three iterations, guided by prior literature and correlation analysis. Iteration 1 used nine features: SMA_10, SMA_50, and SMA_Ratio for trend; RSI14 and MACD for momentum; STDReturns_20 for volatility; Volume_Ratio for volume; and Lag_1_Return and Lag_5_Return for short-term lag effects. All rolling windows were applied strictly to historical observations to avoid look-ahead bias at the feature level.

In Iteration 2, five additional indicators identified repeatedly in related works were added: ROC_14, Williams_R, Stochastic_14, CCI_20, and OBV. As shown in Figure 1, a correlation heatmap was then reproduced and analysed using a multicollinearity threshold of |r| > 0.80. SMA_10 was removed due to near-perfect correlation with SMA_50, RSI_14 was removed due to high correlation with other momentum indicators, and Williams_R was removed due to near-perfect correlation with Stochastic_14. The resulting revised feature set of 11 indicators was used for model training in Iteration 2.

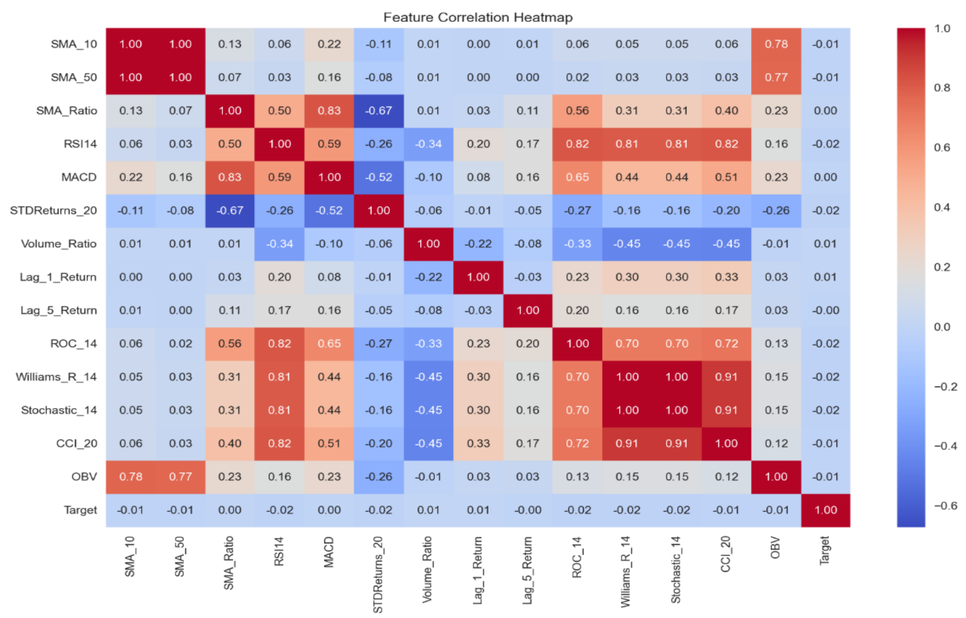

Figure 1: Correlation heatmap produced following Iteration 2 feature engineering, which informed the feature selection decisions described above.

In Iteration 3, STDReturns_20 and Volume_Ratio were systematically removed to isolate the effect of the momentum-only feature configuration on model performance.

This refined momentum-focused configuration produced superior results, suggesting that volatility and volume features introduced noise when combined with denser momentum signals. Feature engineering was implemented in Python using pandas rolling functions and the *ta* library.

3.4 Train/Test Split

A chronological 70/30 split was applied without random shuffling. The training set covered 2005-03-15 to 2018-07-23 (3,362 rows), while the test set covered 2018-07-24 to 2024-04-15 (1,442 rows). Random shuffling would introduce data leakage through look-ahead bias by allowing the model to train on future observations. Figure 2 illustrates the chronological split relative to SPY's full price history, with the training period capturing the 2008 financial crisis and the test period encompassing the COVID-19 shock and post-pandemic recovery. This split provides a realistic out-of-sample evaluation and a deliberate cross-regime test.

Figure 2: SPY closing price 2005-2024 with a chronological 70/30 train/test split. The red dashed line indicates the split date of 2018-07-23, separating the training period (left) from the test period (right).

3.5 Feature Scaling

A StandardScaler was fitted on the training data only and then applied to the test data. Fitting on training data only prevented leakage of future distributional information. Scaling was particularly important for Logistic Regression, which is sensitive to feature magnitude, while tree-based methods are largely scale-invariant. Nevertheless, scaling was applied consistently across all models for comparability.

3.6 Classification Algorithms and Evaluation Metrics

Four classifiers were tested: Logistic Regression, Random Forest, Gradient Boosting, and XGBoost. Logistic Regression used max_iter = 1000. Random Forest used n_estimators = 200 and max_depth = 5. Gradient Boosting and XGBoost each used n_estimators = 200, learning_rate = 0.05, and max_depth = 3. A random_state of 42 was used throughout for reproducibility. Performance was evaluated using accuracy, precision, recall, F1-score, and AUC-ROC. Because both false positives and false negatives can generate trading losses, F1-score and AUC-ROC were treated as the primary metrics, while the 54.4% naive baseline was used as the main accuracy benchmark. Confusion matrices and ROC curves were also produced for each model and iteration to support visual comparison of results.

4. Results and Discussion

All four classifiers produced near-random results on the baseline nine-feature configuration. Accuracy ranged from 47.92% to 48.68%, with all models performing below the naive baseline of 54.4%. AUC-ROC values similarly remained close to random, ranging from 0.4891 to 0.5106. As shown in Figure 3, Gradient Boosting achieved the highest AUC-ROC at 0.5106, making it the only baseline model to show even marginal discriminative ability above chance. Still, this advantage was negligible in practical terms. This result is consistent with the correlation heatmap discussed in Section 3.3, where all engineered features showed near-zero correlation with the target variable. If no individual feature carries a meaningful predictive signal, even more complex classifiers are unlikely to achieve strong out-of-sample performance. In this respect, the baseline findings align closely with Jiao and Jakubowicz (2017), who also reported near-random AUC values when predicting S&P 500 price direction using historical data alone.

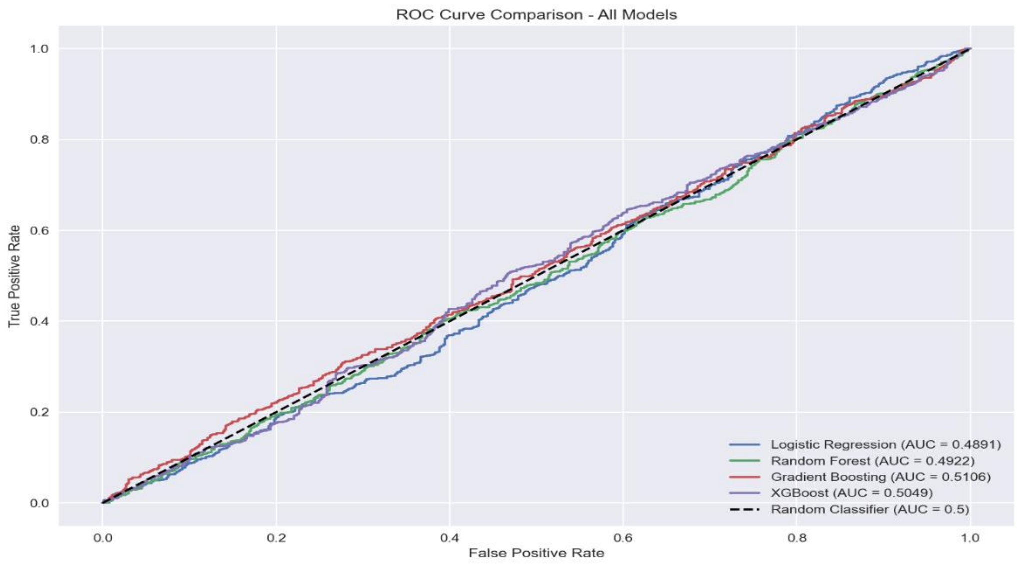

Figure 3: ROC curves for all four classifiers on the baseline nine-feature configuration (Iteration 1), illustrating near-random discriminative ability across all models relative to the random classifier diagonal (AUC = 0.5).

Iterative feature engineering produced only marginal changes, and no model exceeded the naive baseline. As shown in Table 1, the expanded feature set in Iteration 2 led to slight deterioration across most models, contrary to Kumar et al.'s (2018) finding that adding technical indicators improved classification accuracy. The best overall results emerged in Iteration 3, where the refined momentum-focused configuration produced accuracies of 51.39% for Logistic Regression and 51.25% for Gradient Boosting. Even so, both remained below 54.4%. The ROC curves in Figure 4 further reinforce this point, as all models remain close to the random classifier diagonal. The modest improvement in Logistic Regression is likely attributable to the inclusion of bounded oscillator features such as Stochastic and CCI, which provide cleaner and more stable scaled inputs than raw price-based variables. Tree-based models showed little change across iterations, consistent with their scale-invariant nature.

Model	**IT1 Acc**	**IT1 AUC**	**IT2 Acc**	**IT2 AUC**	**IT3 Acc**	**IT3 AUC**
Logistic Regression	0.4868	0.4891	0.4882	0.4863	0.5139	0.4908
Random Forest	0.4792	0.4890	0.4868	0.4890	0.4764	0.4844
Gradient Boosting	0.4820	0.5106	0.4799	0.4936	0.5125	0.4984
XGBoost	0.4827	0.5049	0.4792	0.4863	0.4806	0.4852

Table 1: Model Accuracy and AUC-ROC across three feature configurations. Naive baseline = 54.4%.

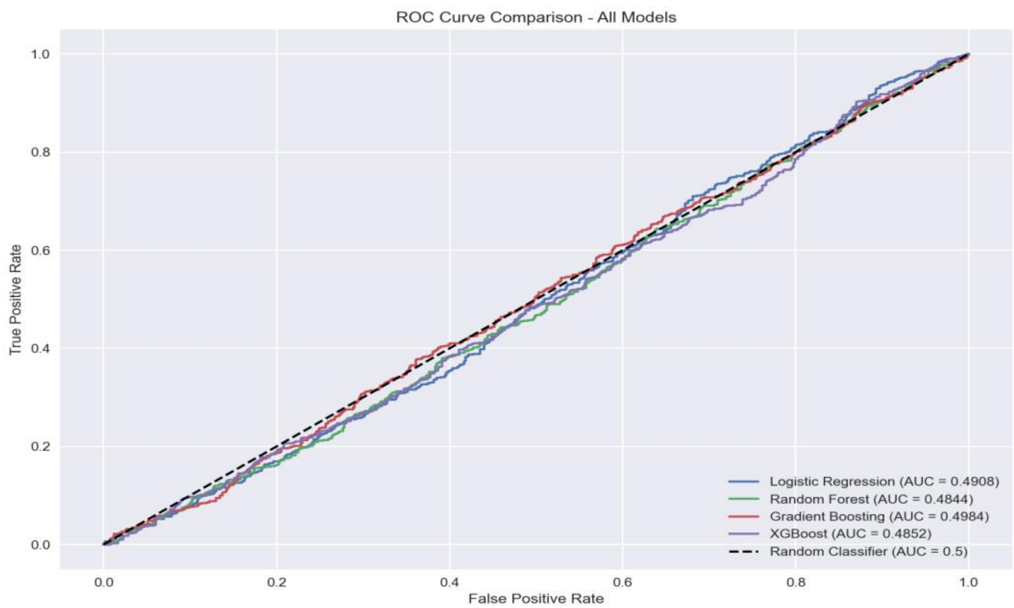

Figure 4: ROC curves for all four classifiers on the refined momentum-focused feature configuration (Iteration 3), demonstrating that AUC-ROC values remained within a narrow band of 0.48 to 0.50 despite iterative feature refinement.

Taken together, the results across all three iterations are broadly consistent with weak-form EMH. AUC-ROC consistently remained within this narrow band across all twelve model-iteration configurations. This pattern is also visible in the confusion matrices shown in Figure 5, which reveal divergent prediction biases across classifiers. This divergence likely reflects the differing sensitivity of linear versus tree-based models to the class imbalance and feature scaling applied, rather than genuine directional insight. The chronological design adds an important further insight. Models trained on data spanning the 2008 financial crisis failed to generalise effectively to the COVID-19 shock and post-pandemic inflationary period, suggesting that crisis regimes are structurally distinct in ways not captured by technical indicators alone. This supports Butler and Kazakov's (2012) argument that market efficiency may vary across regimes. More broadly, the weak results contrast with the 61% to 73% accuracies reported by Kumar et al. (2018), Zouaghia et al. (2023), and Shashank et al. (2023), while differing even more significantly from Bareket and Pârv's (2024) 97% medium-term accuracy. Collectively, this suggests that next-day prediction on a broad market ETF represents a particularly demanding test of weak-form efficiency.

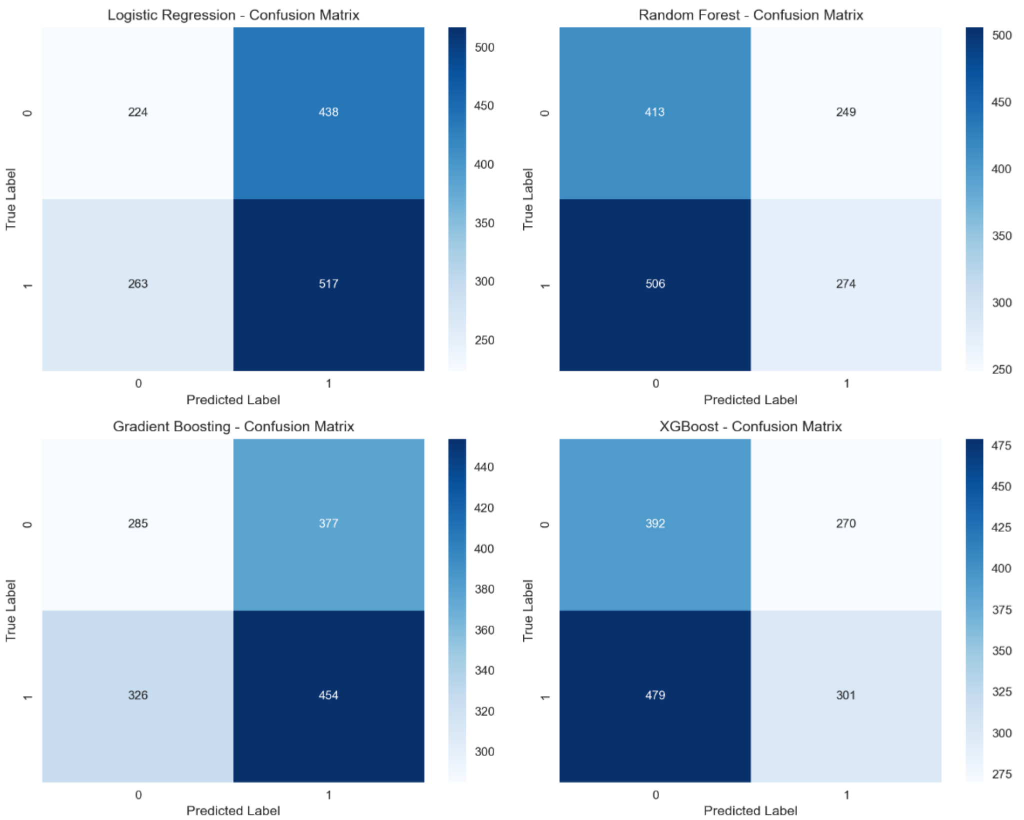

Figure 5: Confusion matrices for all four classifiers on the Iteration 3 feature configuration. Logistic Regression and Gradient Boosting tended toward majority class prediction, generating more false positives than false negatives, while Random Forest and XGBoost showed greater conservatism, producing more false negatives than false positives.

5. Conclusion

This study applied four supervised machine learning classifiers to predict next-day SPY price direction across three iterative feature configurations. Results were consistent across all model-iteration combinations. Taken together, these findings support weak-form EMH, consistent with Fama's (1970) weak-form efficiency framework, suggesting that historical price and volume data alone do not carry sufficient predictive signal for the next-day direction of a highly liquid broad market ETF.

A second key finding is that feature quality mattered more than feature quantity. Expanding the feature set from nine to eleven indicators did not improve performance, while the refined momentum-focused configuration produced the strongest, though still limited, results. This suggests that redundant indicators added noise rather than a meaningful signal, a conclusion reinforced by the multicollinearity analysis undertaken during feature selection.

The study has three main limitations. First, the feature set was restricted to technical indicators and excluded sentiment, macroeconomic, and cross-market variables, which prior studies suggest can improve predictability. Second, evaluation relied on a single chronological split rather than rolling-window validation. Third, the analysis focused only on a fixed next-day prediction horizon. Consequently, the findings may not generalise to other market instruments or longer forecasting windows.

Future research should extend the feature space to include sentiment and macroeconomic data, and test longer prediction horizons. Given Bareket and Pârv's stronger medium-term results, future work should also examine whether predictive power improves when models are evaluated across specific market regime sub-periods rather than a single full-period split, as suggested by Butler and Kazakov's (2012) AMH framework. Ultimately, whether machine learning can consistently exploit market inefficiencies remains an open question, but this study suggests that for a highly efficient market instrument like SPY, the answer lies beyond the reach of historical price data alone.

Supplementary Materials

Three annotated Python scripts and the preprocessed SPY dataset (SPY_dataset.csv) are submitted alongside this report, corresponding to the three feature engineering iterations described in Section 3.3. Iteration 1 contains full inline commentary documenting all preprocessing, feature engineering, modelling, and evaluation decisions. Iterations 2 and 3 follow the same structure, with detailed commentary added for new elements and cross-references to Iteration 1 for unchanged sections. The dataset covers daily SPY OHLCV data from 2005-01-03 to 2024-04-15, sourced from Kaggle. All code was developed and executed in Python 3.14.2 using Visual Studio Code.

References

Akhilesh, P. *et al.* (2024) “Early Forecasting of Stock Marketing Price Using SVM Classifier by Comparing with Decision Tree for Improved Accuracy,” *2024 IEEE 9th International Conference on Engineering Technologies and Applied Sciences (ICETAS)*. *2024 IEEE 9th International Conference on Engineering Technologies and Applied Sciences (ICETAS)*, pp. 1–6. Available at: https://doi.org/10.1109/ICETAS62372.2024.11119925.

Ashtagi, R. *et al.* (2024) “A Comparative Study of Machine Learning Models for Stock Price Prediction,” *2024 2nd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS)*. *2024 2nd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS)*, pp. 768–773. Available at: https://doi.org/10.1109/ICSSAS64001.2024.10760490.

Bareket, A. and Pârv, B. (2024) “Predicting Medium-Term Stock Index Direction Using Constituent Stocks and Machine Learning,” *IEEE Access*, 12, pp. 84968–84983. Available at: https://doi.org/10.1109/ACCESS.2024.3412975.

Batool, K. and Fatima, U. (2025) “Analysis of High Frequency Markets: Global News Impact and Efficient Market Hypothesis Insights,” *2025 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA)*. *2025 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA)*, pp. 1–6. Available at: https://doi.org/10.1109/ACDSA65407.2025.11166630.

Butler, M. and Kazakov, D. (2012) “Testing implications of the Adaptive Market Hypothesis via computational intelligence,” *2012 IEEE Conference on Computational Intelligence for Financial Engineering & Economics (CIFEr)*. *2012 IEEE Conference on Computational Intelligence for Financial Engineering & Economics (CIFEr)*, pp. 1–8. Available at: https://doi.org/10.1109/CIFEr.2012.6327799.

Fama, E.F. (1970) “Efficient Capital Markets: A Review of Theory and Empirical Work,” *The Journal of Finance*, 25(2), pp. 383–417. Available at: https://doi.org/10.2307/2325486.

Jiao, Y. and Jakubowicz, J. (2017) “Predicting stock movement direction with machine learning: An extensive study on S&P 500 stocks,” *2017 IEEE International Conference on Big Data (Big Data)*. *2017 IEEE International Conference on Big Data (Big Data)*, pp. 4705–4713. Available at: https://doi.org/10.1109/BigData.2017.8258518.

Kumar, I. *et al.* (2018) “A Comparative Study of Supervised Machine Learning Algorithms for Stock Market Trend Prediction,” *2018 Second International Conference on Inventive Communication and Computational Technologies (ICICCT)*. *2018 Second International Conference on Inventive Communication and Computational Technologies (ICICCT)*, pp. 1003–1007. Available at: https://doi.org/10.1109/ICICCT.2018.8473214.

Pardeshi, Y.K. and Kale, P. (2021) “Technical Analysis Indicators in Stock Market Using Machine Learning: A Comparative Analysis,” *2021 12th International Conference on Computing Communication and Networking Technologies (ICCCNT)*. *2021 12th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, pp. 1–6. Available at: https://doi.org/10.1109/ICCCNT51525.2021.9580172.

Rouf, N. *et al.* (2021) “Stock Market Prediction Using Machine Learning Techniques: A Decade Survey on Methodologies, Recent Developments, and Future Directions,” *Electronics*, 10(21), p. 2717. Available at: https://doi.org/10.3390/electronics10212717.

Shashank V.N.P. *et al.* (2023) “An Efficient Decision Support System for Stock Market Prediction using Technical Indicators as Features by Machine Learning Approaches,” *2023 4th International Conference for Emerging Technology (INCET)*. *2023 4th International Conference for Emerging Technology (INCET)*, pp. 1–6. Available at: https://doi.org/10.1109/INCET57972.2023.10170137.

Sirohi, A.K., Mahato, P.K. and Attar, V. (2014) “Multiple Kernel Learning for stock price direction prediction,” *2014 International Conference on Advances in Engineering & Technology Research (ICAETR - 2014)*. *2014 International Conference on Advances in Engineering & Technology Research (ICAETR - 2014)*, pp. 1–4. Available at: https://doi.org/10.1109/ICAETR.2014.7012901.

Guillem Servera (2024) *S&P 500, and NASDAQ 100: Daily Data.* Kaggle. Available at: https://www.kaggle.com/datasets/guillemservera/sp500-nasdaq-spy-qqq-ohlcv-data?select=SPY.csv (Accessed: 4 February 2026).

Zouaghia, Z., Kodia Aouina, Z. and Ben Said, L. (2023) “Stock Movement Prediction Based On Technical Indicators Applying Hybrid Machine Learning Models,” *2023 International Symposium on Networks, Computers and Communications (ISNCC)*. *2023 International Symposium on Networks, Computers and Communications (ISNCC)*, pp. 1–4. Available at: https://doi.org/10.1109/ISNCC58260.2023.10323971.

Appendix A: Generative AI Declaration Statement

Declaration Statement

Please tick the appropriate box relating to your use of Gen AI tools for BAA1027: Machine Learning and Advanced Python – Final Assessment.

A. ☐ I did not use any Gen AI tools (including for editing and proofreading) in the completion of this assignment

B. ☒ I did use Gen AI tools in the completion of this assignment

Signature: Dennis Chumakov

Date: 17/04/2026

Appendix B: Reporting of Gen AI Tool Use in Assignment

Tools: I used Gemini (Google) 3.1 Pro between 3rd March 2026 and 15th April 2026.

Outputs: I did not use any direct AI-generated text in this assignment. Gemini was used as an interactive tutor, syntax checker/debugger, and structural feedback tool. All analytical decisions, written content, code logic, and interpretations are my own work.

Example Prompts & Outputs by Category:

1. Conceptual Clarification and Topic Development

   - *Prompt:* Questions about foundational concepts, including the Efficient Market Hypothesis, weak-form efficiency, mean-reversion versus trend persistence, and technical indicators.
   - *Output:* Gemini explained concepts in plain language using real-world examples to support understanding. All claims were independently verified, and the application of these concepts to the research was my own. As I requested, Gemini asked guiding questions to develop my thinking rather than providing answers directly.

2. Research and Source Evaluation

   - *Prompt:* Pasted abstracts from potential IEEE sources and asked for brief yes/no assessments of relevance to my research question.
   - *Output:* Gemini provided short relevance assessments with one-line justifications. All final source selection decisions, detailed reading, and extraction of key findings were conducted independently by me.

3. Python Coding Assistance

   - *Prompt:* Requests for help identifying specific syntax errors and debugging issues encountered while writing code independently in VSCode
   - *Output:* Gemini identified specific errors such as capitalisation mistakes and missing imports. All code was written independently by me based on material covered in tutorials and my past experience with coding. All modelling decisions, parameter choices, and iteration decisions were my own.

4. Structural Guidance

   - *Prompt:* Explained the content I wanted to include in each section, the data and results I had, and the connections to literature. Asked Gemini to structure this into a PPC-style bullet point outline.
   - *Output:* Gemini produced structured outlines based on my input information. All writing was drafted and edited independently by me.

Iterations: Multiple sessions between 3rd March and 15th April 2026, covering conceptual clarification, source evaluation, syntax debugging, and structural guidance as outlined above.

Critical Reflection: I used Gemini responsibly as a tutor and analytical thinking critique rather than a content generator. Gemini's role was to ask guiding questions, explain concepts in plain language, critically evaluate my reasoning and arguments to push my thinking, and provide structural outlines based on my own input. All research decisions, written arguments, feature engineering choices, modelling decisions, critical interpretations, and conclusions remained entirely my own.